

Assignment 1: Requirements Gathering & Specification

Nurshapagat Shapay, Akezhan Kumarov CSE 2501-M

PayBack

Part 1 – Project Planning

PayBack is a web app that allows friends and roommates to track their shared expenses, see who owes what to whom, and pay through Kaspi QR

After a trip or a month of living together, most people don't have a clear understanding of who paid for what. People forget about it or avoid talking about it. PayBack solves this problem with a shared real-time accounting log

The app is mainly used by friends who share travel expenses and roommates who share monthly bills.

- Technically the stack is React, Node.js, and PostgreSQL Kaspi QR is generated client-side with no paid API needed.
- Economically the hosting is free on Vercel and Railway.
- Legally, no real transactions are processed and users own their own data.
- Operationally, the app replaces existing manual habits like notes and group chats with minimal onboarding.
- For the schedule, the team of 2 has 10 weeks MVP is realistic in 4 weeks.

Part 2 – Requirements Documentation

User Stories

To confirm the problem, we interviewed three potential users before drafting the requirements. All three confirmed the core problem:

Question:

In case you have roommates or you go out with your friends on trips how would you share your expenses?

Have you ever been in situations where you have been owed money, but don't know how much so that has not been paid back to you?

Responder 1:

Do you remember our Vietnam trip I paid for everyone? By the end we had no idea who owed how much, too many payments over several days and sadly nobody kept track.

Responder 2:

You know I always log everything in our WhatsApp group chat. It helps, but honestly it's annoying that I get lost in the operations.

Responder 3:

I live alone so really it's not my problem. But the idea itself is good, I'd definitely use it on group trips. In the future I'm planning to visit Tashkent, so maybe I'll use it.

User Stories:

1. US-01 (High) – As a User, I would like to register and log in to make sure that my data will be stored securely.
2. US-02 (High) – As a User, I would like to create a group and invite my friends via the provided link.
3. US-03 (High) – As a User, I would like to create an expense item with the expense name, expense amount, category, and payer.
4. US-04 (High) – As a User, I would like to divide expenses evenly, by percentage, or by a fixed amount.
5. US-05 (High) – As a User, I would like to get automatic calculations regarding who has to pay back who.
6. US-06 (High) – As a User, I would like to simplify the debt repayment process, making less transactions between each other.
7. US-07 (High) – As a User, I would like to get the Kaspi QR-code of a person whom I have debts towards so that I could pay him/her easily.

Non-Functional Requirements

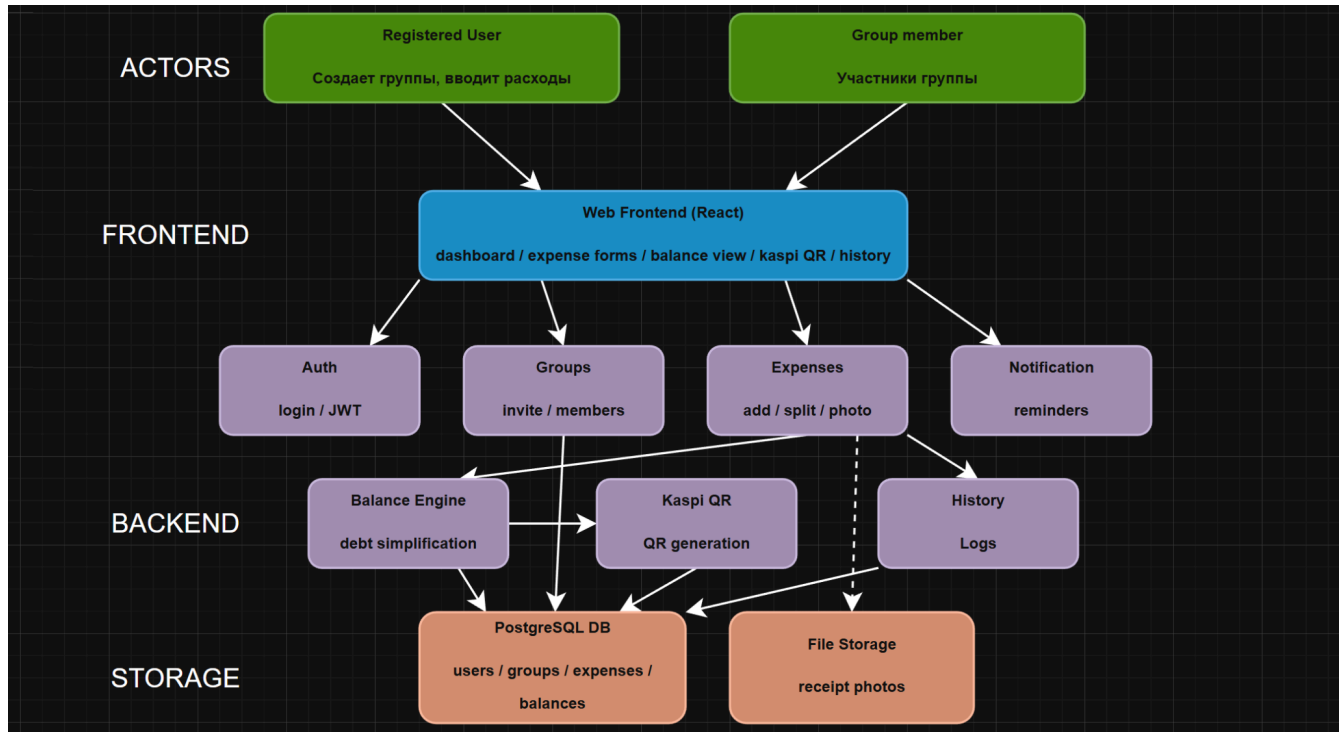
Load time of a mobile application will be under 3 seconds. Recalculation of a balance will be instant and will support up to 300 users. New users are onboard in 5 minutes, without any training, user-friendly UI. Works on mobile devices and as desktop application on personal computers, available in Russian, Kazakh, and English languages.

Passwords must be hashed, HTTPS enforced, sessions expire time 24 hours, users can see only their own groups. Must have 99% uptime, and daily backups. Files over 5MB will be rejected. Stateless backend, database handles 20 members in one group and 500 expenses per group.

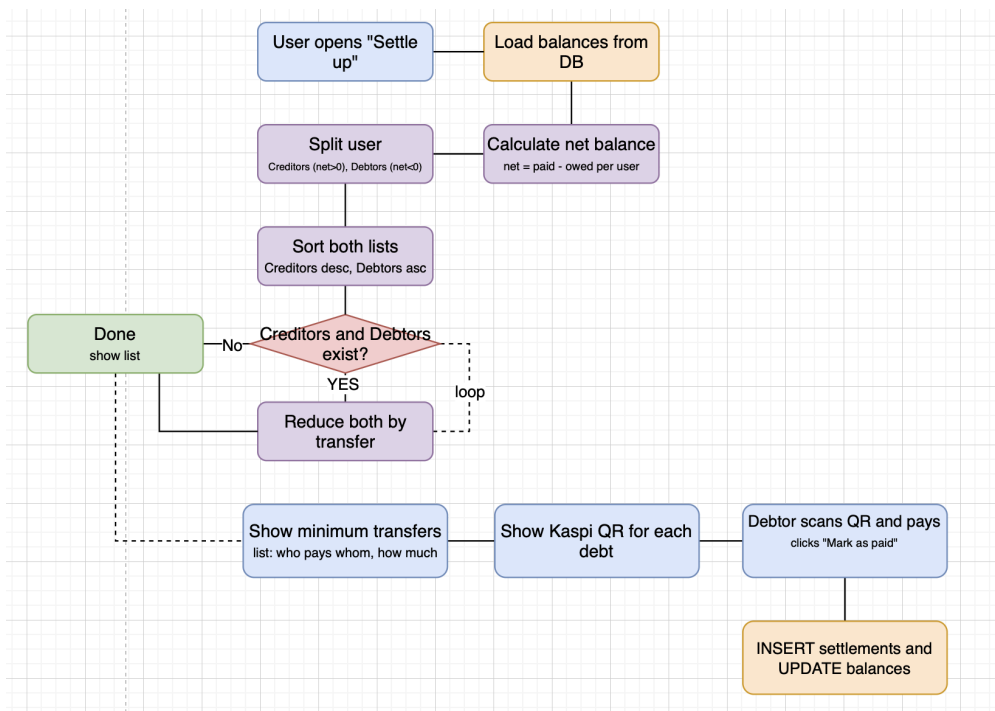
Constraints & Assumptions

Free hosting only, no paid APIs. Kaspi QR generated on client-side. Users assumed to have a Kaspi account and basic digital literacy to work with applications. All amounts are in KZT. The app does not process real payments.

Part 3 - Initial System Scope Diagram



Workflow 1 - Add expenses



Workflow 2 - Debt Simplification

